

Assignment 1: AI System Specification

Option F - Voice-Based Virtual Assistant for Multilingual Customer Support

Course Introduction to Artificial Intelligence	Topic PEAS Framework and Task Environment Analysis	System Type Customer-support conversational agent
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1 PEAS Specification

Performance Measures

- Task resolution rate: customer issues solved without human transfer.
- Language accuracy: correct speech recognition, intent detection, and reply language.
- Customer satisfaction: post-call rating, complaint reduction, and natural tone.
- Average handling time: fast help without rushing or interrupting the customer.
- Escalation quality: complex cases routed with transcript and clear summary.

Environment

A call-center or helpdesk setting where customers speak different languages, use accents, ask about orders, billing, refunds, accounts, or complaints, and may call from noisy places.

Actuators

- Speak answers through text-to-speech in the customer's selected language.
- Ask follow-up questions, confirm details, or repeat unclear information.
- Create tickets, update CRM records, check order status, and trigger support workflows.
- Transfer to a human agent with language, transcript, and issue summary.

Sensors

- Customer audio, microphone quality, background noise, pauses, and interruptions.
- Speech-to-text transcript, language detection, accent cues, and confidence scores.
- Intent, sentiment, urgency, account history, previous tickets, and CRM profile data.
- Knowledge-base articles, product policies, backend systems, and live agent availability.

2 Environment Classification

Dimension	Choice	Justification
Observability	Partially Observable	The assistant cannot directly know the customer's real emotion, full problem, or background situation.
Agents	Multi-agent	The caller, virtual assistant, human agents, backend systems, and compliance tools all affect the result.
Determinism	Stochastic	The same reply can lead to different outcomes because speech quality, mood, accent, and context vary.
Episodes	Sequential	Each question changes the next conversation state, so earlier misunderstandings can affect the whole call.
State Change	Dynamic	Speech, order status, queue length, and agent availability can change while the call is active.
Continuity	Continuous	Speech confidence, sentiment, waiting time, urgency, and risk are measured on continuous scales.

3 Critical Analysis

Greatest technical challenge

The biggest challenge is partial observability. In a real support call, the assistant hears only the customer's voice and sees limited account data. It may miss stress, sarcasm, background pressure, or the real reason behind a complaint. Multilingual support makes this harder because accents, code-switching, slang, and noisy audio can confuse speech recognition. A good engineer needs confidence checks and safe handoff rules, otherwise the system may sound fluent while solving the wrong problem.

Utility function and weight change

A useful utility function is: $U = 0.30 \cdot \text{resolution} + 0.25 \cdot \text{language_accuracy} + 0.15 \cdot \text{customer_satisfaction} + 0.10 \cdot \text{speed} - 0.15 \cdot \text{escalation_cost} - 0.25 \cdot \text{misinformation_risk}$. If the misinformation-risk weight is doubled to 0.50, the assistant becomes more careful. It asks more confirmation questions and transfers uncertain billing, legal, or account-security cases to a human earlier. Calls may become a little slower, but the chance of harmful or incorrect support is reduced.

Valid JSON object for the PEAS and environment analysis:

```
{
  "system_name": "Voice-Based Virtual Assistant for Multilingual Customer Support",
  "peas": {
    "performance_measures": [
      "task_resolution_rate",
      "language_accuracy",
      "customer_satisfaction",
      "average_handling_time",
      "escalation_quality"
    ],
    "environment": "Multilingual helpdesk with callers, CRM, policies, and backend systems.",
    "actuators": [
      "spoken_replies",
      "follow_up_questions",
      "crm_updates",
      "ticket_creation",
      "workflow_triggering",
      "human_agent_transfer"
    ],
    "sensors": [
      "customer_audio",
      "speech_to_text",
      "language_detection",
      "intent_detection",
      "sentiment_score",
      "crm_history",
      "knowledge_base",
      "agent_availability"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "Intent and emotion are inferred from speech and account signals."
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "Customers, support agents, and backend systems interact."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Speech quality and customer reactions are uncertain."
    },
    "episodic_sequential": {
      "choice": "Sequential",
      "justification": "Each response affects the next turn."
    },
    "static_dynamic": {
      "choice": "Dynamic",
      "justification": "Caller, queue, and records can change during the call."
    },
    "discrete_continuous": {
      "choice": "Continuous",
      "justification": "Confidence, sentiment, waiting time, and risk use continuous scores."
    }
  },
  "utility_function": "U=0.30R+0.25L+0.15S+0.10T-0.15E-0.25M"
}
```